Advancing Brain-Computer Interface (BCI) Technology

Background and Motivation

I am currently a Master's student in Microelectronics Technology and Materials at The Hong Kong Polytechnic University. Before pursuing my Master's degree, I accumulated over ten years of professional experience in developing image processing algorithms and AI-based image recognition solutions. I previously worked as an Algorithm Engineer and R&D Lead at notable organizations, including the publicly listed company OPT and the prominent AI firm SmartMore Technology.

My skill set includes programming, engineering debugging, setting up and deploying integrated development environments, data collection and preprocessing, parameter optimization, model training and testing, result analysis and refinement, as well as replicating and enhancing cutting-edge technologies. I have successfully filed and been granted more than ten invention patents. Additionally, I possess expertise in designing Bluetooth communication devices, drawing circuit schematics, and creating 3D models and printed components. My prior projects include using AI technology to control multi-axis robotic arms and other complex systems.

I have long been interested in Brain-Computer Interface (BCI) technology, which I regard as a guiding vision for my career development. To me, BCI is a multidisciplinary field that requires a balanced focus: 50% on engineering and computer science (including AI and electronics), 20% on materials science and microphysics, and 30% on bioengineering and medical science (especially neuroscience and brain science). Over the past decade, I have concentrated on software development, image processing, and AI applications while maintaining a strong passion for computer and electronic technologies. With the rapid advancements in BCI, I believe now is the perfect time to take a decisive step forward into this field.

Current Focus and Future Plans

To transition into BCI research, I have revisited foundational technologies, such as cochlear implants and OpenBCI systems, while reorganizing my technical expertise. With a solid background in image processing, software development (mainly in C++ and Python), and AI applications, I decided to pursue a Master's degree in Semiconductor Materials and Technology to address my knowledge gaps in nanomaterials and microphysics. These interdisciplinary skills are critical for improving the precision and accuracy of signal acquisition in BCI, where electrode materials and their interactions with biological signals are particularly important.

Looking ahead, I plan to complement my engineering expertise with foundational knowledge in biology, neuroscience, and medical science, enabling me to collaborate more effectively with experts in BCI teams. My current research focus will continue to center on software development, signal acquisition and processing, and model training and optimization. By analyzing large datasets of neural signals, I aim to identify patterns and use AI techniques to enhance signal recognition accuracy. Moreover, I am particularly interested in exploring how brain signals can control robotic arms, combining my prior experience to develop better rehabilitation solutions for patients.

Research Goals

If given the opportunity to pursue a PhD, I aim to achieve the following research objectives:

- **Data Acquisition and Analysis**: Collect and preprocess large volumes of neural signal data, ensuring its quality through meticulous cleaning, classification, and validation.
- **Model Development**: Design and refine brain signal recognition models, leveraging AI technology to improve accuracy, robustness, and stability.
- **Real-World Applications**: Investigate the feasibility of brain-controlled devices, particularly in integrating BCI technology with robotics.
- **Interdisciplinary Collaboration**: Combine software engineering with neuroscience, brain science, and materials science to deliver innovative BCI solutions, especially focusing on advancements in electrode technology.

Through these efforts, I aspire to contribute to the development and clinical validation of BCI technologies, ultimately making a meaningful societal impact.

Challenges and Preparedness

I fully recognize the challenges in this field, such as low initial recognition rates for models, stability issues, and difficulties in transferring results across devices or environments. However, my extensive professional experience equips me well to address these challenges. I understand the critical role of data quality in determining the success of a model and how improvements in sensor precision, signal processing techniques, and experimental setups can significantly enhance research outcomes.

Furthermore, I am aware of the importance of interdisciplinary collaboration. For instance, in BCI, electrode materials play a decisive role in signal acquisition accuracy and reliability, and understanding their properties and interactions with biological systems is crucial. Similarly, collaborating with medical professionals requires a solid grasp of medical protocols and effective communication. By equipping myself with knowledge in neuroscience, materials science, and engineering, I am preparing methodically to overcome these challenges.

Commitment to BCI and the Greater Bay Area

I have deep ties to the Greater Bay Area, with my family based in Nansha District, Guangzhou, and eight years of professional experience in Shenzhen, including four years as an entrepreneur. I am fluent in Cantonese, Mandarin, and English (with basic proficiency in Japanese), which enables me to thrive in the region's international innovation ecosystem.

The Greater Bay Area provides a unique environment for advancing BCI technology, offering world-class medical facilities, abundant data resources, and a vibrant culture of innovation. I am committed to establishing myself in this region, channeling my research findings back into the local community to drive technological development and enhance social welfare.

Conclusion

Thank you for taking the time to review my research proposal. I am eager to contribute to the advancement of BCI technology and look forward to the opportunity to collaborate with like-minded researchers. Please feel free to contact me if you have any questions or need additional information.